

EXECUTIVE SUMMARY

Although Social Media platforms are now of the main channels for exchanging information, they have also hastened the spread of False information and deceptive content. Because contemporary misinformation frequently blends textual assertions with alerted or deceptive content, it is difficult to identify fake news in such settings. As a result, traditional text – only detection techniques frequently lack trustworthy procedures for verifying claims against outside evidence and struggle to identify multimodal contradictions. In order to detect fake news, this idea suggest an Evidence – based multimodal approach that incorporates external knowledge verification, image – text consistency analysis and textual semantics. To access the semantic congruence between news claims and its related photos, the system first uses congruence embeddings created by Sentence -BERT to represent textual information. Visual features then processed by CLIP-Based encoder. A retrieval-based verification module is developed to further evaluated the information’s credibility. Using FAISS Vector search and SBERT Embeddings, pertinent evidence is retrieved from an external Knowledge source. A Natural Language Inference (NLI) model based on BART-Large-MNLI is then used to analyze the recovered evidence in order to ascertain whether the claim is supported, refuted, or unrelated to the material at hand. A feature Fusion process is used to merge the features obtained from the text semantics, Multimodal consistency, and Evidence verification, and LightGBM model is used for classification. The Suggested framework performs well across common classification criteria. The reliability of detecting fake news is further enhanced by combining these multimodal signals with retrieved- based evidence verification.

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List of Abbreviations

TF-IDF	Term Frequency-Inverse Document Frequency
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term memory
BERT	Bidirectional Encoder Representations from Transformers
SBERT	Sentence -BERT
NLP	Natural Language Processing
CLIP	Contrastive Language-Image Pre-Training
FAISS	Facebook AI Similarity Search
NLI	Natural Language Inference
BART-Large-	Bidirectional and Auto-Regressive Transformers-Large-Multi-
MNLI	Genre Natural Language Inference
LLM	Large Language Model
RAG	Retrieval Augmented Generation
URL	Uniform Resource Locator
SVM	Support Vector Machine
CNN	Convolutional Neural Network
LightGBM	Light Gradient Boosting machine
AI	Artificial Intelligence
API	Application Programming Interface
GPU	Graphics Processing Unit
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative

Symbols and Notations

$f(I)$	Image Embedding.
$g(T)$	Textual Embedding.
$f_k(x)$	Predction of Kth decision tree.
$Text_{768}$	SBERT Embeddings – 768 Dimensions.
$CLIP_1$	CLIP score – 1 Dimension.
RAG_1	RAG-NLI score – 1d Dimension.
$P_{contradiction}$	Probability assigned by NLI model indicating that the evidence contradicts the claim.
$P_{entailment}$	Probability assigned by NLI model indicating that the evidence supports (entails) the claim.